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Summary

Alimentary canal		
Organs	**Functions**	**Secretions**
Mouth	Receive food, starts mechanical digestion by chewing with the help of teeth and tongue.	—
Pharynx	Swallowing	—
Esophagus	conduction of the food to stomach	—
Stomach	Stores and churns food. Initiates protein digestion. Limited absorption.	Mucus protects stomach. HCL activates enzymes and kills germs. Pepsin digests protein. Gastrin hormone stimulates HCl and pepsinogen secretion.
Small Intestine	Completes digestion and absorbs nutrients	Mucus protects gut wall. Peptidases digest proteins. Maltase, lactase and sucrase digest sugars. Lipase digests lipids. Nuclease digests nucleic acids. Cholecystokinin triggers the release of pancreatic juices and bile from the gallbladder.
Large Intestine	Reabsorbs water, ions, vitamins. Stores wastes.	—
Rectum	Expels wastes	—
Anus	Opening for waste elimination	—

Accessory digestive glands		
Glands	**Functions**	**Secretions**
Salivary Glands	Moistens food	Salivary mucus lubricates food. Salivary amylase (Ptyalin) digests starch.
Liver	Produces Bile, synthesises cholesterol and steroids. Destroys old blood cells. Detoxifies poisons produced by bacteria and others in food. Breaks heamoglobin of dead and damaged RBC and convert into bile pigments.	Bile emulsifies fat
Gall Bladder	Stores bile	
Pancreas	Exocrine part secretes digestive enzymes. Endocrine part secretes hormones such as insulin by beta cells, glucagon by alpha cells which regulates blood glucose levels.	Bicarbonate neutralizes stomach acid. Trypsin and chymotrypsin digest proteins. Carboxypeptidase digests peptides. Amylase digests starch and glycogen. Lipase digests lipids. Nuclease digests nucleic acids.